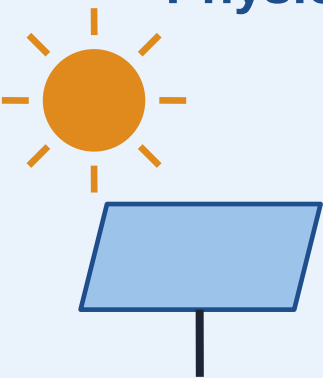


## Physical PV data



**Yulara 768 kW**

off-grid, desert

**NIST 271 kW**

grid-connected, humid

**Curtailed series removed**

≈50% of its energy curtailed



**Archived GFS forecasts**

only runs published in time

## Honest selection (AHSE)

LightGBM

XGBoost

PatchTST

N-HiTS

LSTM

GRU

+ GFS inputs for tree models

per lead-time block: 1 h, 1–6 h, 6–24 h

**combine models only if the  
bootstrap gain is > 0**

otherwise: best single model

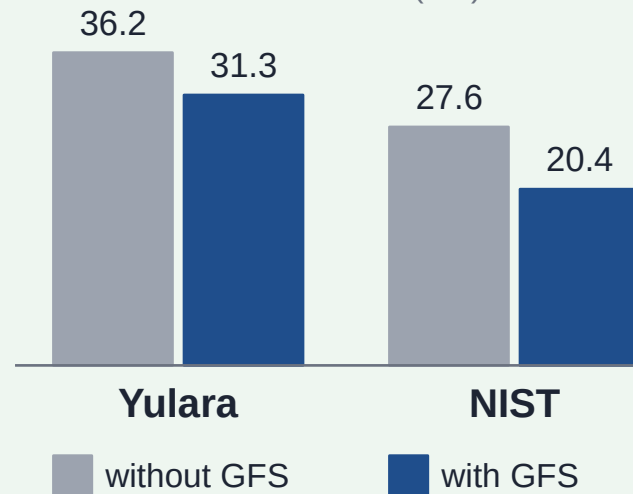
**Decided on validation data only**

+ out-of-fold conformal 90% intervals

## Test results, 1–24 h ahead

**GFS lowers MAE: –13% (Yulara), –26% (NIST)**

AHSE test MAE (kW)



**90% intervals: coverage 0.90–0.92**

AHSE never significantly worse than best model